

Make Your Own JOYSTICK FOR ROBOT

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Here is a simple joystick to control a robot to move in forward, backward, right, left, clockwise (CW) and counter-clockwise

(CCW) directions using six buttons. It is cheaper than most other joysticks available in the market. The author's prototype is shown in Fig. 1.

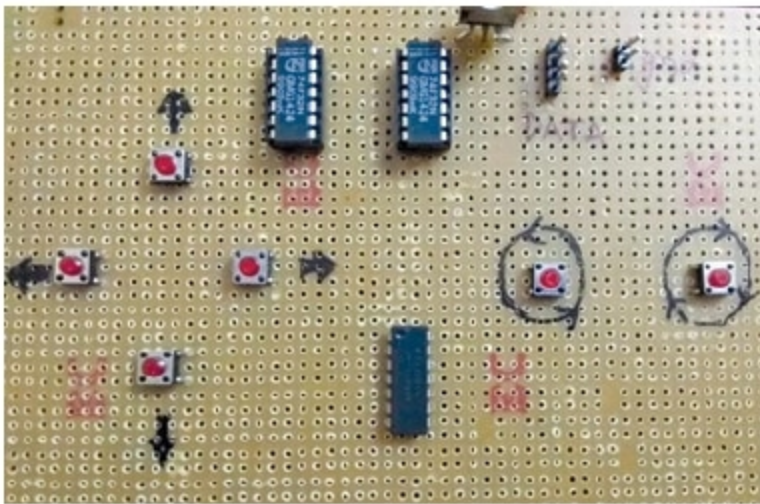


Fig. 1: Author's prototype

Circuit and working

Circuit diagram of the joystick for the robot is shown in Fig. 2. It is built around two quad OR gates 7432 (IC1 and IC2), hex inverter 7404 (IC3), four LEDs, six tactile switches (S1 through S6) and a few other components.

Logic for each move, like forward, backward, left, right, CW and CCW

rotations, is based on the truth table using 6x4 encoder circuit formed by IC1, IC2 and IC3 logic gates. On pressing any one of the six switches (S1 through S6), a corresponding binary-coded decimal (BCD) code is generated as per the truth table.

When input to OR gate goes high, its output is high. Corresponding output is connected to input pin of NOT gate (hex inverter). When input to NOT gate is high, its output is low. Same is the case with other inputs.

All input lines of OR gates are pulled down to ground using resistors R1 through R6. When forward switch S1 is pressed, output of OR gate N1 of IC1 goes high, which makes N2 of IC1 (OR gate) go high. At the same time, it makes OR gate N3 of IC1 go high, which makes N1 of IC2 (OR gate) go high.

Outputs of IC1 and IC2 are fed to input pins 1 and 5 of IC3 (NOT gate), respectively. Before feeding the signals to IC3, code generated is 1010. But after passing through IC3, output becomes 0101. This is the required logic for moving the robot in forward direction as per the truth table.

Similarly, when backward switch S4 is pressed, it makes output gate N4 of IC1 and output gate N2 of IC2 go high. Before the signals reach IC3, code is 0101, and after passing through IC3, it becomes 1010. This is the required logic for moving the robot in backward direction as per the truth table.

Output codes generated for all input switches are available at connector CON2.

Construction and testing

An actual-size PCB layout of the joystick for the robot is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect a 5V DC across CON1.

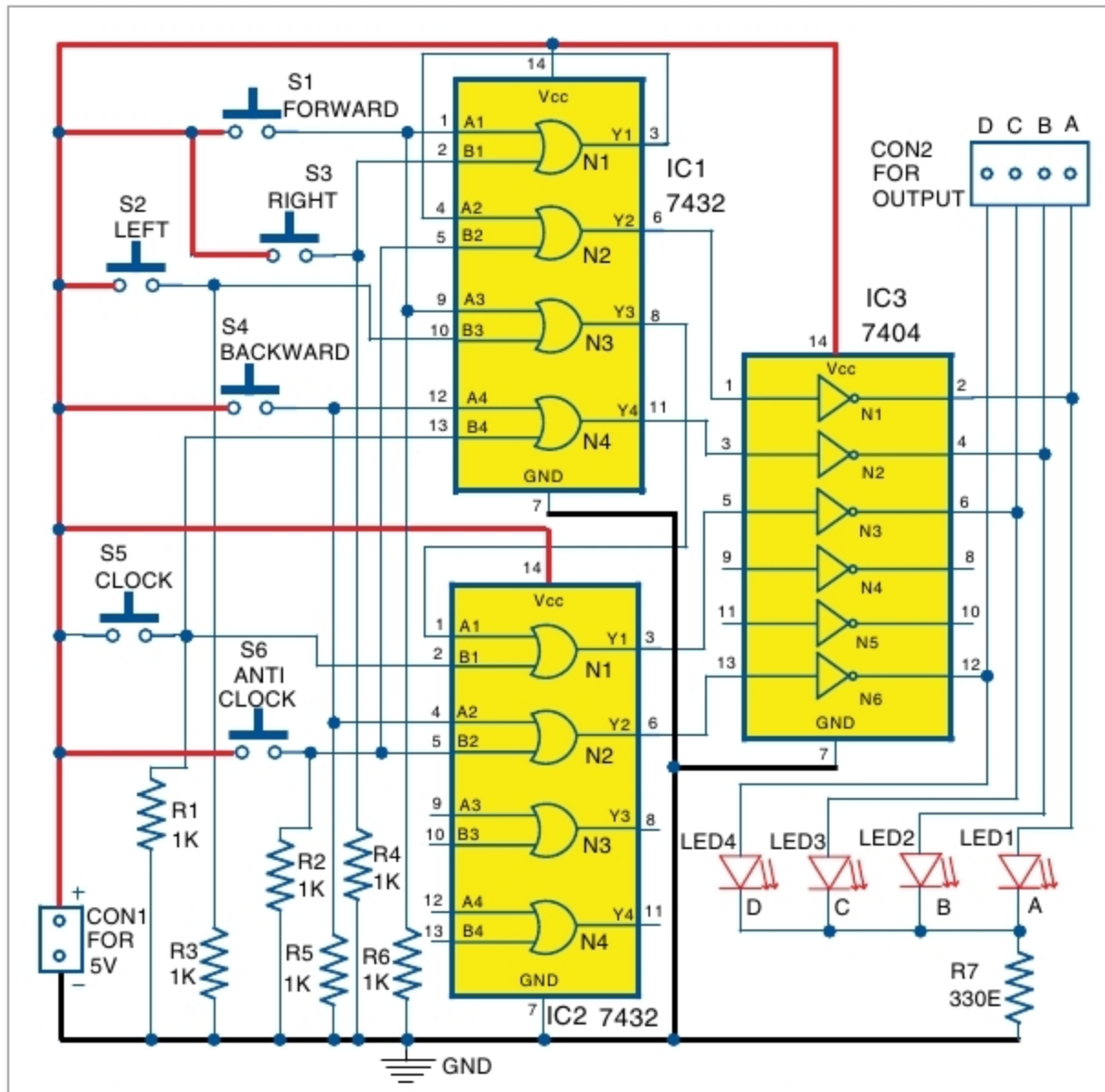


Fig. 2: Circuit diagram of joystick for robot